

BHAVAN JASANI

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SUMMARY

Machine learning and computer vision scientist with 6+ years of applied research experience building and deploying large-scale transformer-based large language models for (1) multimodal learning — across images, text, video, audio, and structured data (2) synthetic data generation & annotation — with humans and AI in the loop to overcome scarce or hard-to-label data. Published in top computer vision and machine learning conferences (CVPR, ECCV, ICCV), mentored PhD interns, filed patents, and collaborated with cross-functional teams to translate research ideas into production-grade models solving business problems.

SKILLS

- **Programming:** Python, PyTorch, TensorFlow, DeepSpeed, FSDP, Hugging Face, PySpark, JAX, Gradio, C/C++
- **ML & Deep Learning:** Pre-training, SFT, RLHF & Preference Optimization, LoRA/Adapters, Vision-Language & Multimodal Models, Vision Transformers, Diffusion Models, Self-Supervised Learning, Human-in-the-Loop Data & Annotation Pipelines, Synthetic Data Generation
- **Bio-ML:** AlphaGenome, AlphaFold, Evo, ESM, STATE Virtual Cell Model
- **Systems:** Large-Scale Distributed Training, AWS (EC2/S3/Lambda), Docker, Kubernetes, Git

EXPERIENCE

Amazon Web Services (AWS) AI

San Francisco, CA

Applied Scientist, AWS AI labs (Computer Vision)

September 2019 – present

- Designed and deployed multimodal transformer architectures for document understanding, launching Amazon Textract Queries, the first commercial document VQA system (related research published at ICCV)
- Built large-scale pre-training, fine-tuning, and preference learning pipelines integrating spatial, textual, and visual modalities for VQA and document models
- Developed synthetic data generation pipelines combining expert humans, non-expert annotators, tools, and vision-language models (published at CVPR)
- Led end-to-end development of a multimodal AI assistant for documents, images, video, and audio as part of Amazon Bedrock Data Automation
- Created visual grounding methods for model explainability, linking language-model predictions to corresponding image regions
- Mentored PhD interns, collaborated cross-functionally, published in CVPR/ICCV/ECCV, and filed patents on multimodal transformer architectures

Carnegie Mellon University, Robotics Institute, School of Computer Science

Pittsburgh, PA

Research Assistant

October 2017 – August 2019

- Built multimodal emotion recognition models integrating audio, video, facial action units, body pose, and 3D facial landmarks for clinical psychology datasets
- Contributed to an NIH-funded study identifying behavioral biomarkers of depression using HIPAA-compliant psychotherapy session videos, collaborating with an interdisciplinary team of psychologists, clinicians, and computer scientists

Nanyang Technological University, School of Computer Science & Engineering

Singapore

Research Staff

January 2016 – May 2017

- Implemented a parallel and hardware-efficient DPM object detection algorithm for real-time pedestrian detection on FPGA-based embedded system, achieving a 40% reduction in hardware resources usage
- Developed a bit-width optimization approach for hardware acceleration of the Harris Corner Detector algorithm, achieving 335 FPS on HD video with minimal accuracy loss

SELECTED PUBLICATIONS

- **Synthesize Step-by-Step: Tools, Templates and LLMs as Data Generators for Reasoning-Based Chart VQA**, *CVPR 2024*
- **DocFormer: End-to-End Transformer for Document Understanding**, *ICCV 2021*
- **YORO - Lightweight End-to-End Visual Grounding**, *ECCV Workshops 2022*
- **Are We Asking the Right Questions in MovieQA?**, *ICCV Workshops 2019 (spotlight oral)*
- **Exploiting Spatial Layout in Document Question Answering using Transformers**, *AMLC 2021*
- **End-to-End Visual Question Answering on Document Images**, *AMLC 2021*
- **Threshold-Guided Design and Optimization for Harris Corner Detector Architecture**, *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) 2017*
- **Skeleton-Based Zero-Shot Action Recognition in Joint Pose-Language Semantic Space**, *arXiv 2019*
- **Automatic Detection of Human Affective Behavior in Dyadic Conversations**, *Master's Thesis, Robotics Institute, Carnegie Mellon University 2019*

PATENTS

- **Global Prompts with Linear Adapter Tuning for Regression-Free Model Update**, *US patent filed 2023*
- **Document Visual Question Answering with Multimodal Transformer Encoder-Decoder Models**, *US patent filed 2022*

COMMUNITY SERVICE

- **Reviewer** for CVPR, ICCV, ECCV, AMLC, ACVC and Amazon Research Awards
- **Program Committee Member** for ICDAR 2025 and TASK-CV Workshop at ICCV 2019
- **Book Chapter Reviewer** for Data Augmentation with Python, Packt publishing 2023

AWARDS & ACHIEVEMENTS

- **EB1-B (Outstanding Researcher) Green Card**, granted for demonstrated scientific impact by U.S. Government
- **Research Assistantship**, Carnegie Mellon University (2017 – 2019)
- **DAAD WISE scholarship**, German Academic Exchange Service (2014)
- **INSPIRE Fellowship**, Department of Science & Technology, Government of India (2011 – 2016)

EDUCATION

- **Carnegie Mellon University, School of Computer Science** 2017 – 2019
M.S. in Robotics (Research based – Computer Vision & Machine Learning)
- **Birla Institute of Technology & Science (BITS), Pilani – K.K. Birla Goa** 2011 – 2016
M.Sc. Physics
- **Birla Institute of Technology & Science (BITS), Pilani – K.K. Birla Goa** 2011 – 2016
B.E. Electrical & Electronics Engineering